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HCI Quiz 2 solution

1) Given Parameters data:

- Starting Position S: (533, 650)
- Button A (Bounding Box):

Corner (412, 541), width (W_A) = 53
Height (H_A) = 30

• Button B:

Corner (616, 288)

width (W_B) = 64

Height (H_B) = 29

• Button C:

Corner (477, 592)

width (W_C) = 61

Height (H_C) = 24

Fitts law Constants:

$a = 0.25$ (reaction time)

$b = 0.165$ (pointing/device constant)

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2 - Calculation of Target Center Coordinates:

$$\text{Center} = \left(\text{corner}_x + \frac{\text{Width}}{2}, \text{corner}_y + \frac{\text{Height}}{2} \right)$$

Center of Button A (A_c):

$$= \left(412 + \frac{53}{2}, 541 + \frac{30}{2} \right)$$

$$\Rightarrow (438.5, 566.0)$$

Center of Button B (B_c):

$$\left(616 + \frac{64}{2}, 288 + \frac{29}{2} \right)$$

$$\Rightarrow (616 + 32, 288 + 14.5)$$

$$\Rightarrow (648.0, 302.5)$$

Center of Button C (C_c):

$$\left(477 + \frac{1}{2}, 592 + \frac{24}{2} \right)$$

$$\Rightarrow (477.5, 604.0)$$

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3. Calculation of pointing distances

• S to center of A (A_{SA}):

$$\sqrt{(438.5 - 533)^2 + (556 - 650)^2} \\ \approx 133.290 \text{ px}$$

• S to center of B (A_{SB}):

$$\sqrt{(448 - 533)^2 + (302.5 - 650)^2} \\ \approx 366.034 \text{ px}$$

• S to center of C (A_{SC}):

$$\sqrt{(477.5 - 533)^2 + (604 - 650)^2} \\ \approx 72.085 \text{ px}$$

• Center of A to center of B (A_{AB}):

$$\sqrt{(477 - 438.5)^2 + (604 - 556)^2} \\ = \underline{\underline{61.847 \text{ px}}} \\ = 328.8 \text{ px}$$

• Center of A to center of C (A_{AC}):

$$\sqrt{(477.5 - 438.5)^2 + (604 - 556)^2} \\ = 61.847 \text{ px}$$

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• center of B to center of C (ABC)

$$\sqrt{(477.5 - 648)^2 + (604 - 302.5)^2}$$

$$= \sqrt{29070.25 + 90902.22}$$

$$\approx 346.370 \text{ px}$$

4 - Target Acquisition Time (MT):
Sequence 1: ABC

• Step 1 (S → A)

$$ID = \log_2 \left(\frac{2 \times 133 \cdot 290}{53} \right) = 0.5729 \text{ s}$$

• Step 2 (A → B)

$$ID = \log_2 \left(\frac{2 \times 328 \cdot 865}{64} \right) = 0.7378 \text{ s}$$

• Step 3 (B → C)

$$ID = \log_2 \left(\frac{2 \times 346.370}{1} \right) = 1.7098 \text{ s}$$

$$\begin{aligned} \text{Total time (ABC)} &= 0.5729 + 0.7378 + 1.7098 \\ &= 3.0205 \text{ seconds} \end{aligned}$$

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Sequence 2: ACB

Step 1: (S → A)

$$ID = \log_2 \left(\frac{2 \times 61.847}{1} \right) = 1.3121s$$

Step 2: (A → C)

$$ID = \log_2 \left(\frac{2 \times 61.847}{1} \right) = 1.3121s$$

Step 3: (C → B)

$$ID = \log_2 \left(\frac{2 \times 346.370}{64} \right) = 0.7498s$$

Total Time (ACB) = 2.6348 seconds.

Sequence 3: BAC

Step 1: (S → B)

$$ID = \log_2 \left(\frac{2 \times 366.034}{34} \right) = 0.7625s$$

Step 2: (B → A):

$$ID = \log_2 \left(\frac{2 \times 328.864}{53} \right) = 0.7801s$$

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(A → C)

$$MT_3 = 1.3121s$$

$$\text{Total time} = 2.8560 \text{ seconds}$$

Sequence 4 (BCA)

(S → B)

$$MT_1 = 0.765s$$

(B → C)

$$MT_2 = 1.7098s$$

(C → A)

$$ID = \log_2 \left(\frac{2 \times 66.847}{53} \right) = 0.3956$$

$$\text{Total Time} = 2.8680 \text{ seconds}$$

Sequence 5 : LAB

(S → A)

$$MT_1 = 1.3475s$$

(C → A)

$$MT_2 = 0.3956s$$

(A → B)

$$MT_3 = 0.7378s$$

$$\text{Total Time} = 2.4809 \text{ seconds.}$$

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Sequence 6 : CBA

(S → C)

$$MT_1 = 1.3475s$$

(C → B)

$$MT_2 = 0.7498s$$

(B → A)

$$MT_3 = 0.7813s$$

$$\text{Total Time} = \cancel{0.7813s}$$

$$2.8786 \text{ seconds.}$$

5 - Comparative Analysis Summary

Sequence Path	Time (c=0)
ABC	2.6595s
ACB	2.2749s
BAC	2.4514s
BCA	2.5687s
CAB	<u>2.1880s</u>
CBA	2.4754s

6 - Conclusion:

Sequence (CAB) takes the least amount of time of complete.

Optimal execution time: 2.48 seconds.